NAGESH MANSUKE

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PROFESSIONAL SUMMARY

Final-year Computer Science Engineering student and full-stack developer with production experience across the MERN stack, Java, and Spring Boot. Built LoadStoreCounter, an out-of-tree LLVM analysis pass in C++17 (New Pass Manager) that statically counts memory load/store instructions across a module, packaged as a CMake plugin and verified against LLVM 18. Strong foundation in Data Structures & Algorithms (200+ LeetCode problems solved), REST API design, and authentication/authorization systems. Seeking to contribute to open-source compiler and systems projects through the Linux Foundation LFX Mentorship Program.

EDUCATION

Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded *2023 – 2027*

Bachelor of Technology in Computer Science Engineering

TECHNICAL SKILLS

- Programming Languages: Java, JavaScript, Python, C, C++17
- Compilers & Systems Programming: LLVM 17+, LLVM IR, New Pass Manager, Module/Function Pass Development, Clang, opt, CMake (out-of-tree plugins)
- Frontend Development: HTML5, CSS3, React.js, Tailwind CSS, Bootstrap
- Backend Development: Node.js, Express.js, RESTful API Design
- Databases: MongoDB
- Developer Tools: Git, GitHub, VS Code, Postman, CMake
- Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, JWT Authentication, Role-Based Access Control (RBAC)

WORK EXPERIENCE

Web Development Intern — Solutions Touch *June 2026 – Present*

Official College Website

- Contribute to the development and maintenance of the official website of Shri Guru Gobind Singhji Institute of Engineering and Technology, developing responsive modules with React.js, Node.js, Express.js, and MongoDB.
- Build and test RESTful APIs, collaborating through Git, GitHub, and Postman while following industry-standard software development practices.

KEY PROJECTS

LoadStoreCounter — Out-of-Tree LLVM Analysis Pass

C++17, LLVM 17+ (New Pass Manager), CMake, Clang, Bash

- Designed and implemented a read-only LLVM Module Pass in C++17 that traverses Module → Function → BasicBlock → Instruction, classifying LoadInst/StoreInst via isa<> to report per-function and module-wide memory-access counts.
- Built on LLVM's New Pass Manager (PassInfoMixin, PreservedAnalyses) and packaged as a dynamically loadable CMake plugin (add_llvm_pass_plugin), registering a custom pipeline (load-store-counter) via llvmGetPassPluginInfo for use with opt -load-pass-plugin.
- Handled real-world IR edge cases (e.g., skipping declaration-only functions like external printf) and preserved all downstream analyses by returning PreservedAnalyses::all(), since the pass performs zero IR mutation.
- Authored an automated Bash pipeline (run.sh) that builds the plugin, compiles a companion C test program to LLVM IR via Clang, and runs the full pass end-to-end; verified against LLVM 18 with a complete technical report and README.

EduHive — Smart Collaborative Learning Platform

React.js, Node.js, Express.js, MongoDB, AI Integration

- Built a full-stack AI-powered collaborative learning platform for students and teachers using the MERN stack, with JWT-based authentication and Role-Based Access Control across Admin, Teacher, and Student roles.
- Integrated AI study assistance, responsive dashboards, and real-time collaboration features.
- Designed scalable REST APIs and optimized MongoDB database operations for improved application performance.

TALK-ORBIT — Real-Time Chat Application

React.js, Node.js, Express.js, MongoDB, Socket.IO

- Developed a real-time chat application using Socket.IO with one-to-one and group messaging, secure authentication, and persistent chat history in MongoDB.
- Designed modular REST APIs and an event-driven backend architecture for efficient, low-latency communication.

ACHIEVEMENTS & LEADERSHIP

- Solved 200+ Data Structures and Algorithms problems on LeetCode; selected among the Top 10 teams in HackFusion 2.0, a state-level hackathon.
- Team Member, Official College Web Team — collaborate with developers to maintain and enhance institutional web applications.
- Joint Coordinator, college cultural fest "UTSAV" — coordinated event planning and cross-functional team collaboration.